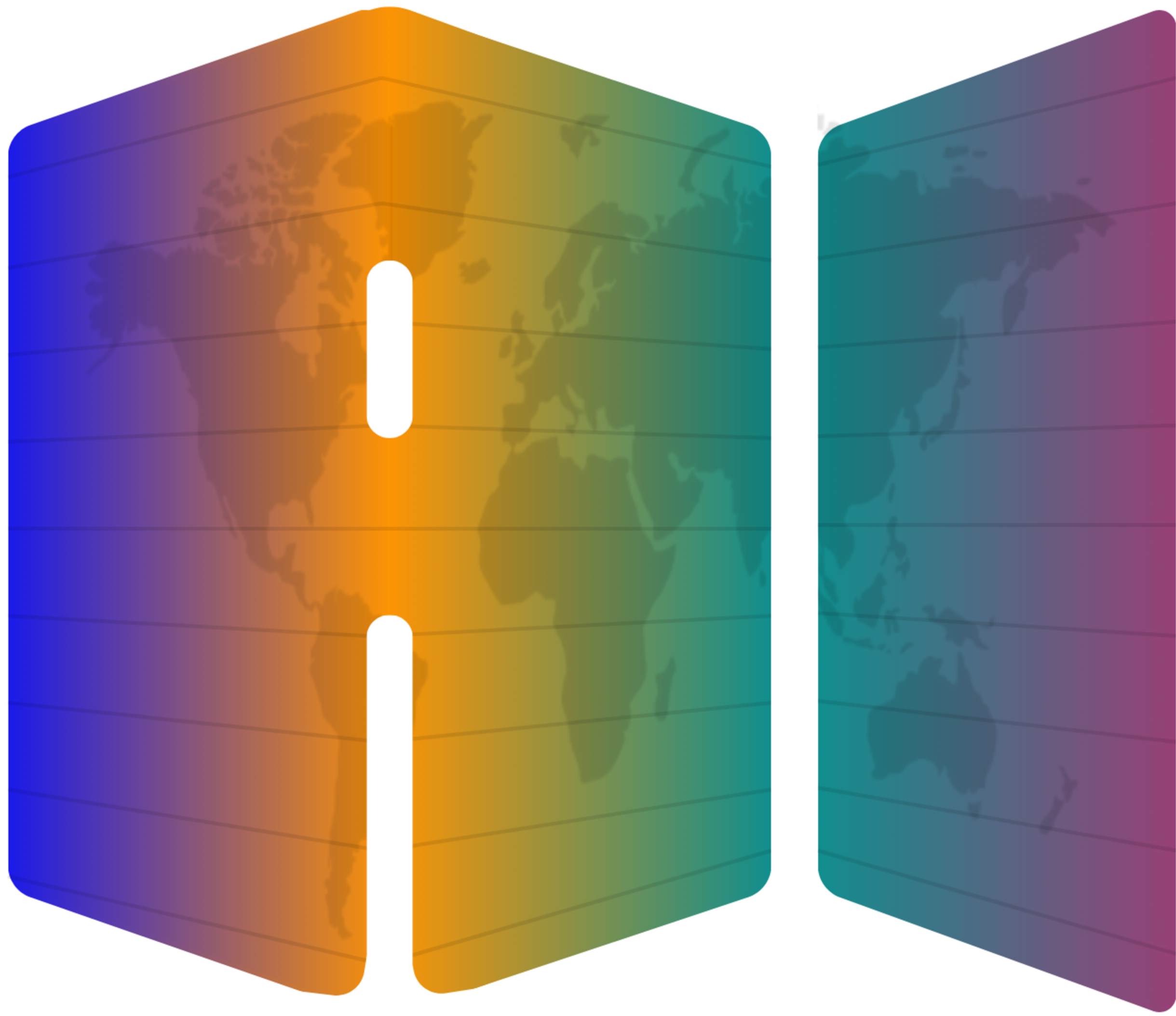
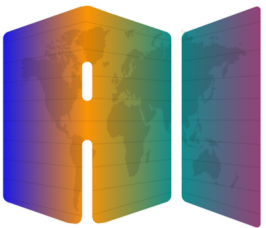




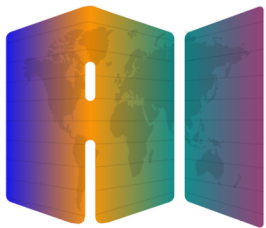
Agentic AI

Lectures by Andrew Ng



Practical Tips for Building AI Agents

Evaluations (evals)

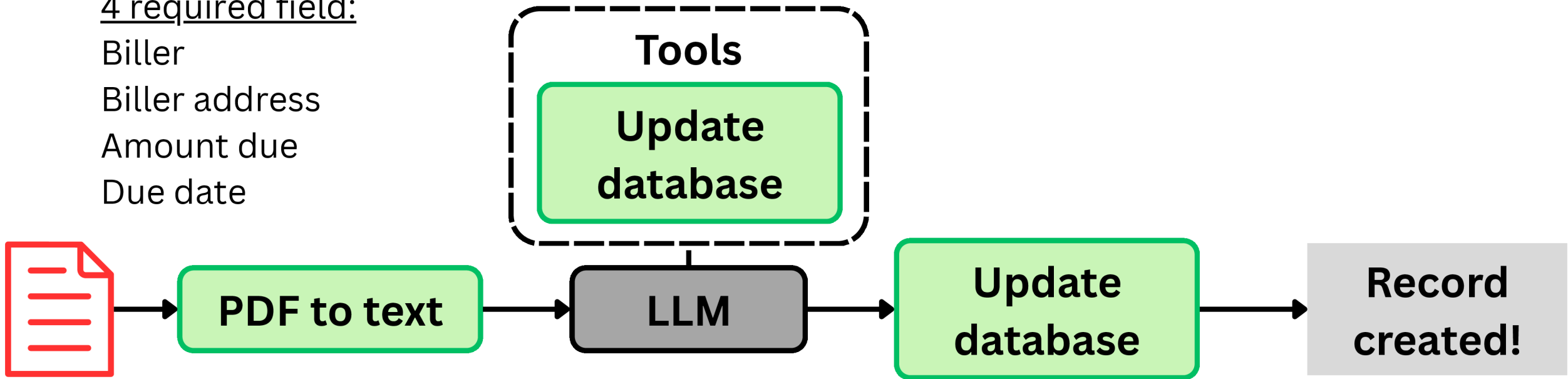


Example: Invoice processing workflow

TechFlow Solutions LLC 8085 Mangga Street Paranaque, NCR 1700 Phone: 500-5361 Email: billing@techflowsol.com			
Due Date: August 20,2025		Invoice Date: August 6, 2025	
Description	Qty	Unit Price	Line Total
Consulting - Systems Integration (hrs)	20	P6000	P120000
Total Due:			P120000

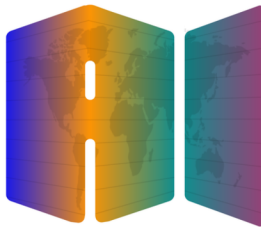
4 required field:

Biller
Biller address
Amount due
Due date



10-20 invoices

Invoice 1	Fine
Invoice 2	Mixed up dates
Invoice 3	Fine
...	...
Invoice 19	Mixed up dates
Invoice 20	Mixed up dates



Create an eval to measure date extraction

1. Manually extract due dates from 10-20 invoices

test invoice 1



“August 20, 2025”



“2025/08/20”

per example ground truth

2. Specify output format of data in prompt

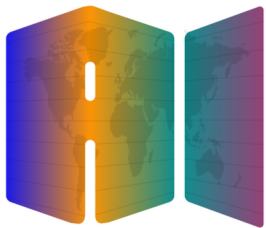
*Format the due date as
YYYY/MM/DD*

3. Extract date from the LLM response using code

```
date_pattern = r'\d{4}/\d{2}/\d{2}'  
extracted_date = re.findall(date_pattern, llm_response)
```

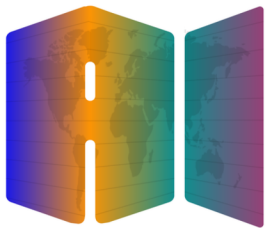
4. Compare LLM result to ground truth

```
if (extracted_date == actual_date):  
    num_correct += 1
```



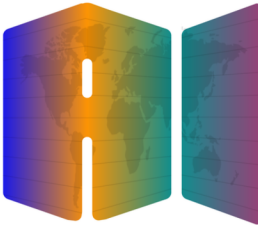
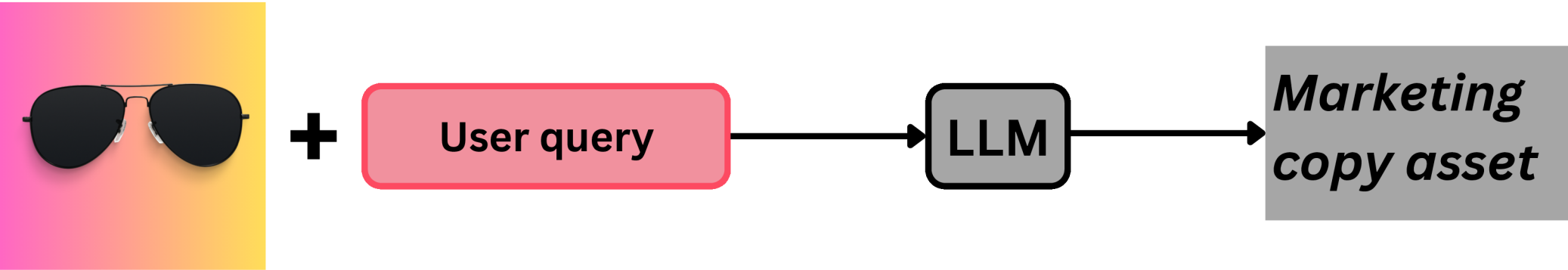
Create an eval to measure date extraction

- Build a system and look at outputs to discover where it is behaving in an unsatisfactory way
 - e.g. incorrect due dates in invoice data extract
- Drive improvement by putting in place a small eval with ~20 examples to help you track progress
- Monitor as you make changes to workflow (e.g. new prompts, new algorithms) and see if the metric improves



Example: marketing copy assistant

Length guidelines:
Instagram caption: 10 words max



	17 words
	Ok
	Ok
	14 words
	11 words

Create an eval to measure text length

1. Create a set of 10-20 test tasks

Image

Example Prompt



Create an Instagram post



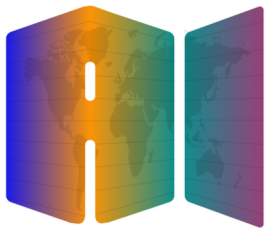
Create an Instagram post

2. Add code to measure word count of the output

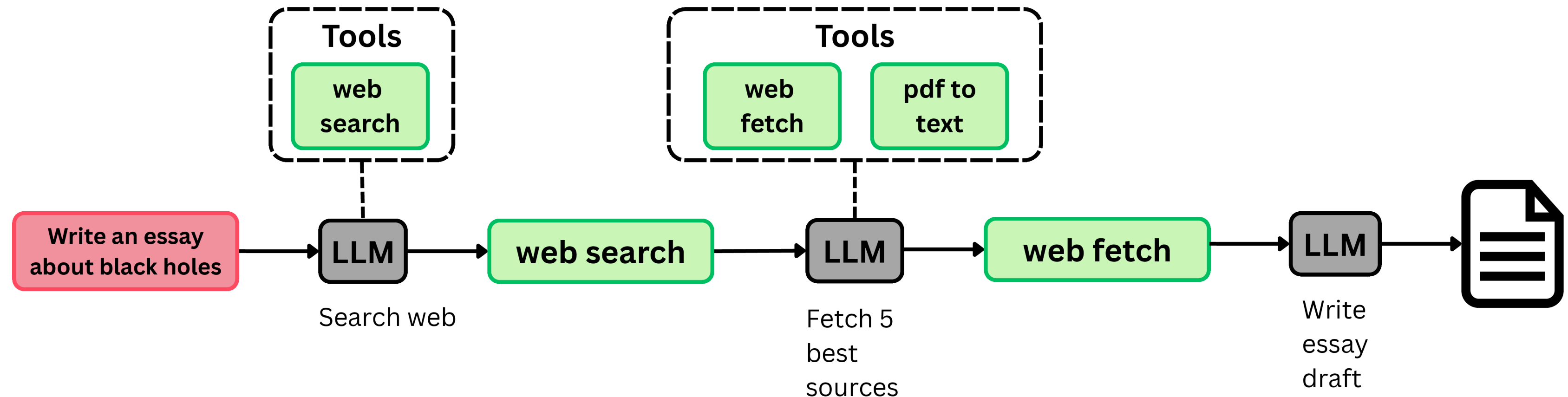
```
word_count = len(text.split())
```

3. Compare length of generated text to limit

```
if (word_count <= 10):  
    num_correct += 1
```

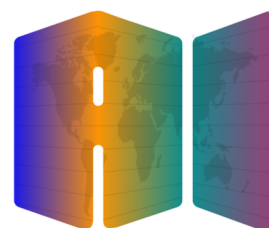


Example: research agent



Propmt	Issues
Recent black hole science	Missed high-profile result that had lots of news coverage
Renting vs buying a home in Seattle?	Seems to do a good job
Robotics for harvesting fruit	Didn't mention leading equipment company

Sometimes misses points a human would have made



Create an eval to measure performance

1. Choose 3-5 gold standard discussion points for each topic

Example prompt	Gold-standard talking points
Black holes	Event horizon, radio telescope
Robotic harvesting	RoboPick, pinchers

ground truth annotations

2. Use LLM-as-a-judge to count how many topics were mentioned

3. Get score for each prompt in eval set

Determine how many of the 5 gold-standard talking points are present in the provided essay.

Original Prompt

{original_prompt}

Essay to Evaluate

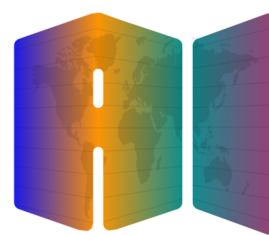
{essay_text}

Gold Standard Talking Points

{gold_standard_points}

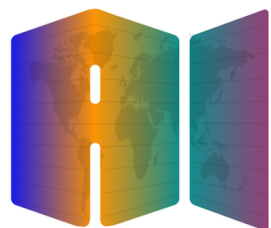
Output Format

Return a json object with two keys: score (a single number between 0 and 5), and explanation (a string that lists the talking points present)



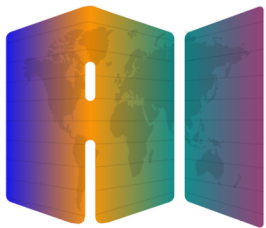
Two “axes” of evaluation

	Evaluate with code (objective)	LLM-as-judge (subjective)
Per example ground truth	Checking invoice date extraction <pre>if (extracted_date == actual_date): num_correct += 1</pre>	Counting gold-standard talking points Count the number of gold standard points in the following text...
No per example ground truth	Checking marketing copy length <pre>if len(text) <= 10: num_correct += 1</pre>	Grading charts with a rubric Grade this chart according to i) whether it has clear axes labels, ii)...



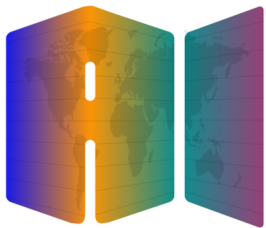
Tips for designing end-to-end evals

- Quick and dirty is ok to start!
- As you find places where your evals fail to capture human judgement as to what system is better, use that as an opportunity to improve the metric
- Look for places where performance is worse than humans

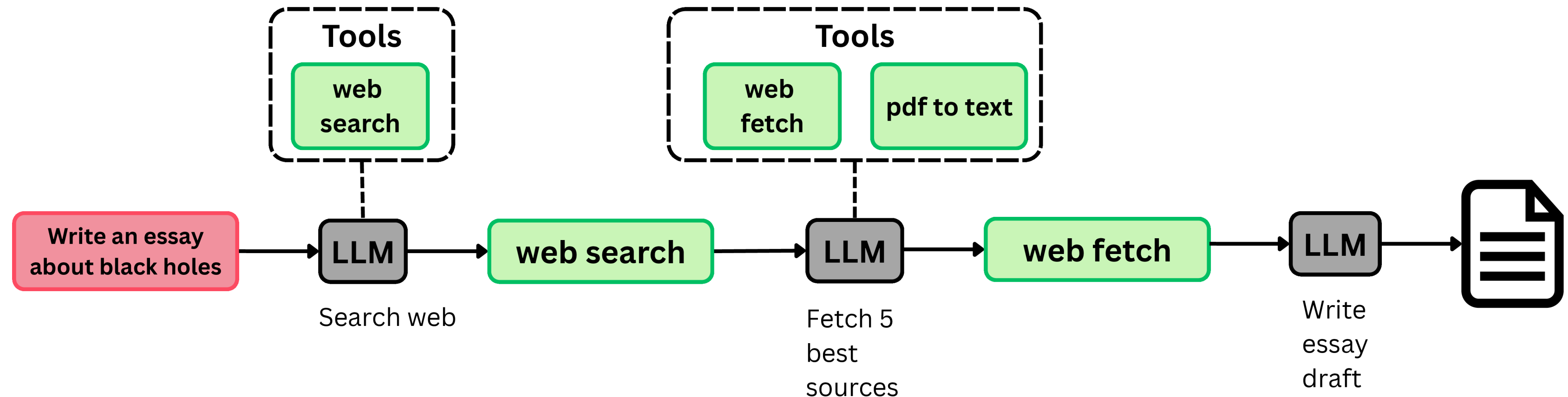


Practical Tips for Building AI Agents

**Error analysis and prioritizing
next steps**

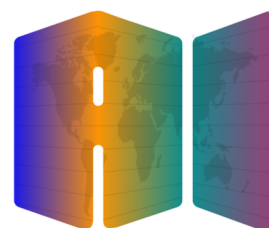


Example: research agent

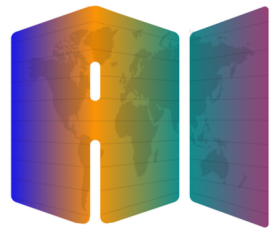
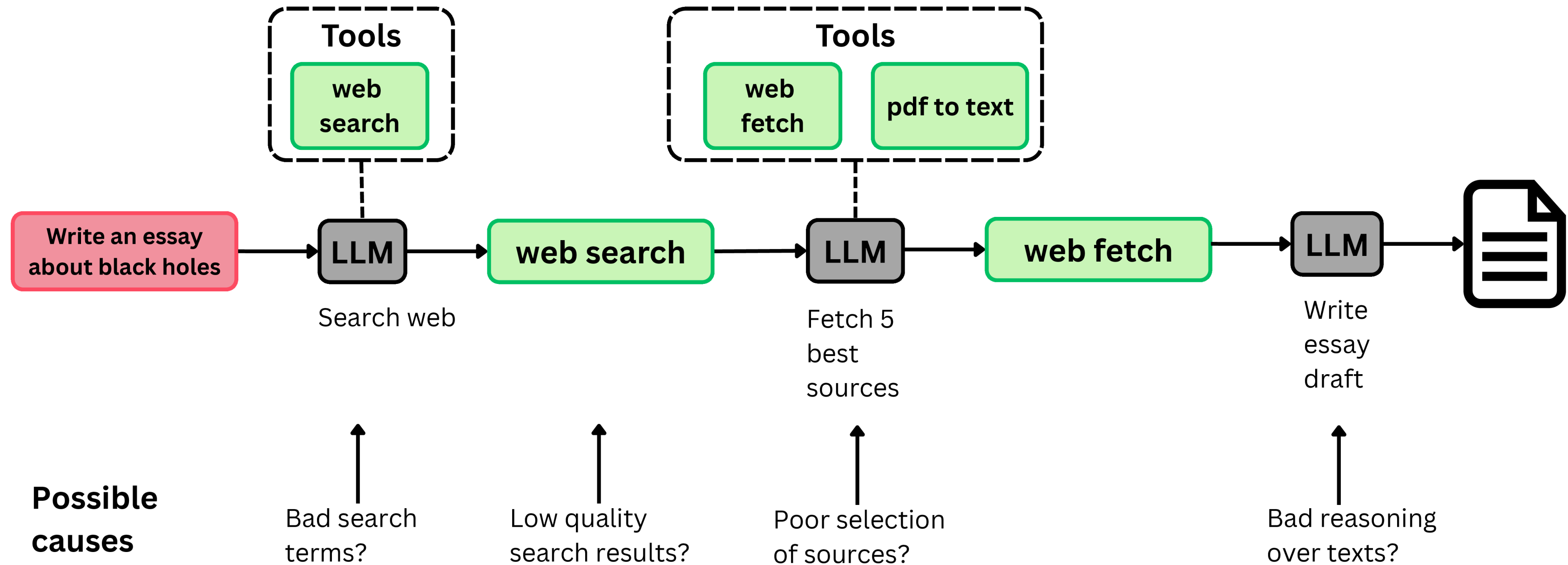


Propmt	Issues
Recent black hole science	Missed high-profile result that had lots of news coverage
Renting vs buying a home in Seattle?	Seems to do a good job
Robotics for harvesting fruit	Didn't mention leading equipment company

Sometimes misses points a human would have made

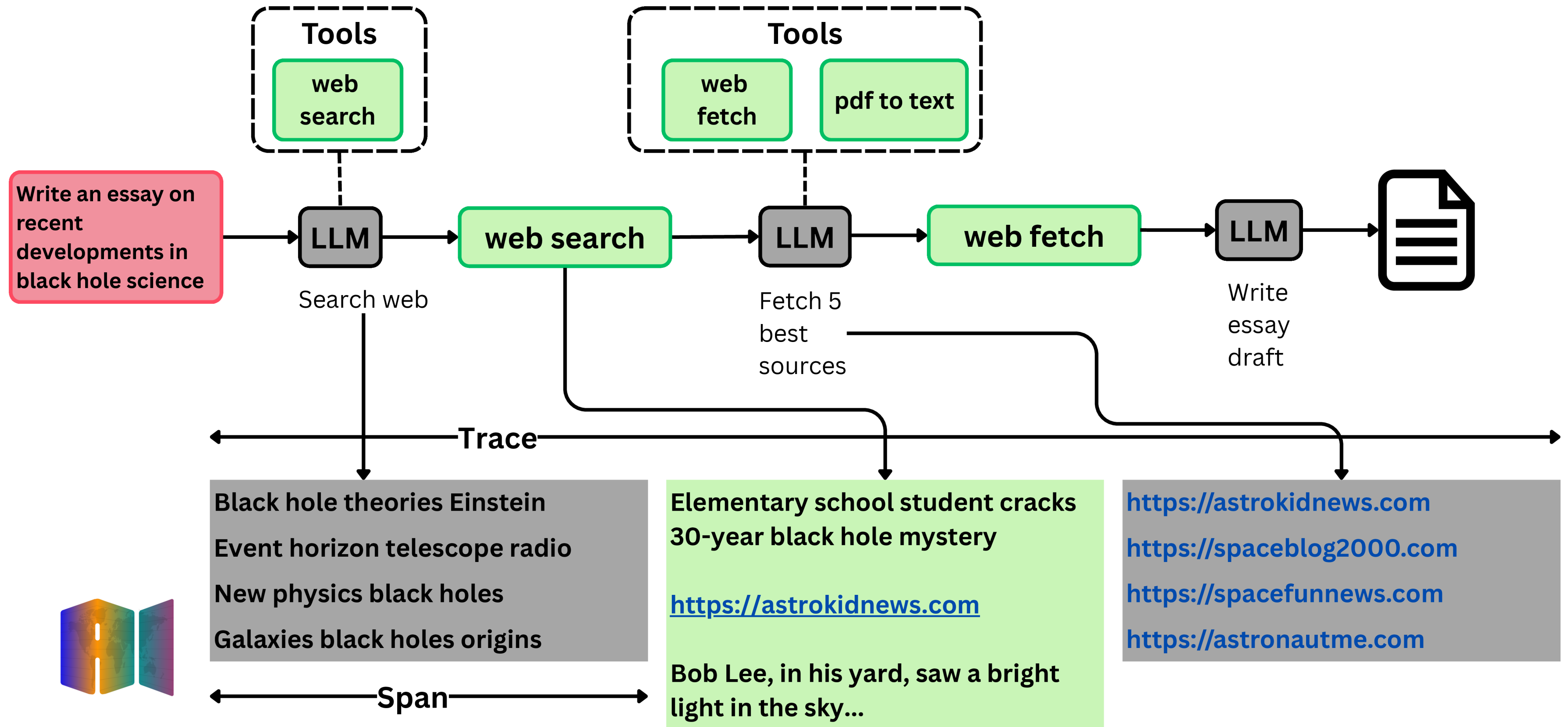


Example: research agent



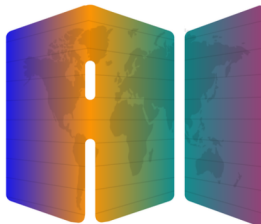
Examine traces to better understand each step in the workflow

Looking at traces



Counting up the errors

Prompt	Search terms	Search results	Picking 5 best sources	...	...
Recent developments in black hole science		Too many blog posts, not enough papers			
Renting vs buying a home in Seattle			Missed well-known blog		
Robotics for harvesting fruit	Terms too generic	Website for elementary school students			
...	...	...	...		
Batteries for electric vehicles		Only selected US-based companies	Missed magazine		



5%

45%

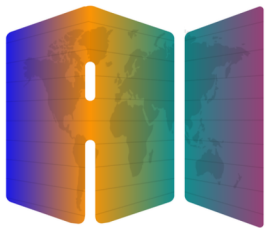
10%

...

...

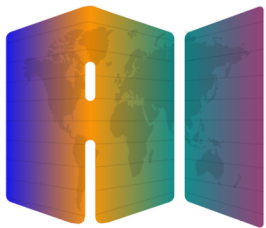
Tips for error analysis

- **Develop a habit of looking at traces**
- **Carry out error analysis to figure out what component performed poorly, leading to a poor final output**
- **Use error analysis output to decide where to focus efforts**



Practical Tips for Building AI Agents

More error analysis examples



Example: Invoice processing workflow

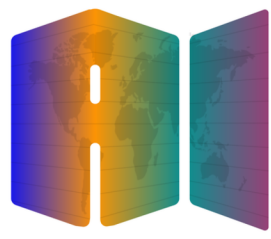
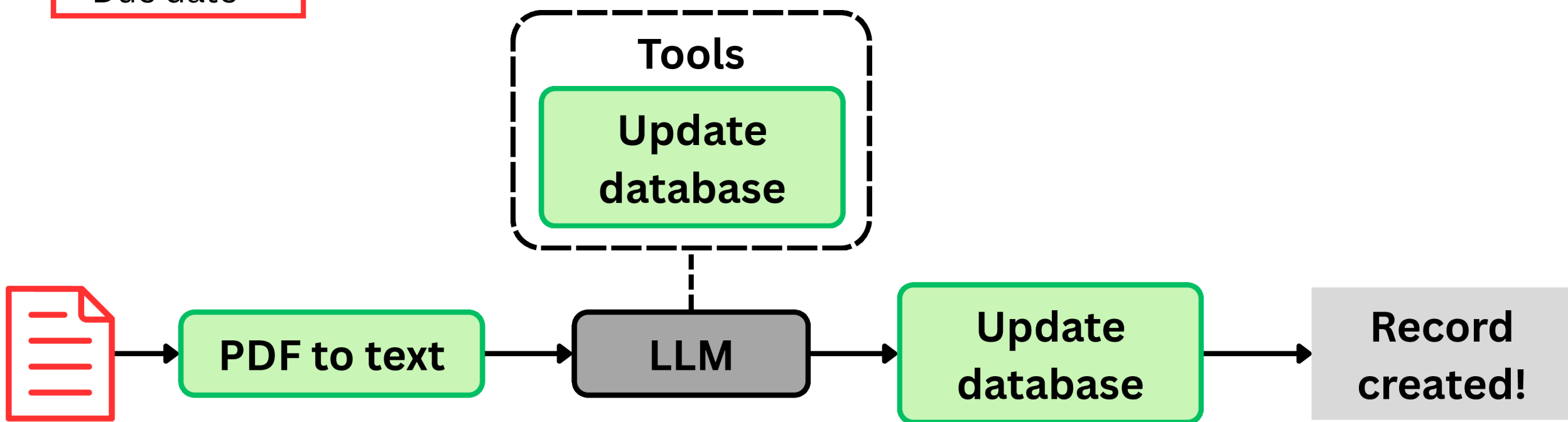
TechFlow Solutions LLC 8085 Mangga Street Paranaque, NCR 1700 Phone: 500-5361 Email: billing@techflowsol.com			
Due Date: August 20,2025		Invoice Date: August 6, 2025	
Description	Qty	Unit Price	Line Total
Consulting - Systems Integration (hrs)	20	P6000	P120000
Total Due:			P120000

4 required field:

- Biller
- Biller address
- Amount due
- Due date

Steps:

- 1. Identify required fields
- 2. Record in database



To carry out error analysis, focus on examples where performance is subpar

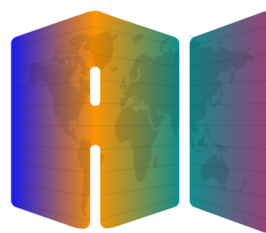
Counting up the errors

- Select 10-100 invoices for which the agentic workflow extracted the wrong due date

Input	PDF-to-text	LLM data extraction
Invoice 1	Errors in extraction	
Invoice 2		Wrong date selected
Invoice 3		Wrong date selected
...	...	...
Invoice 20	Errors in extraction	Wrong date selected

15%

87%



Example: Responding to customer email

From: Susan Jones

Subject: Wrong item shipped

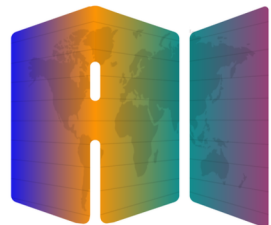
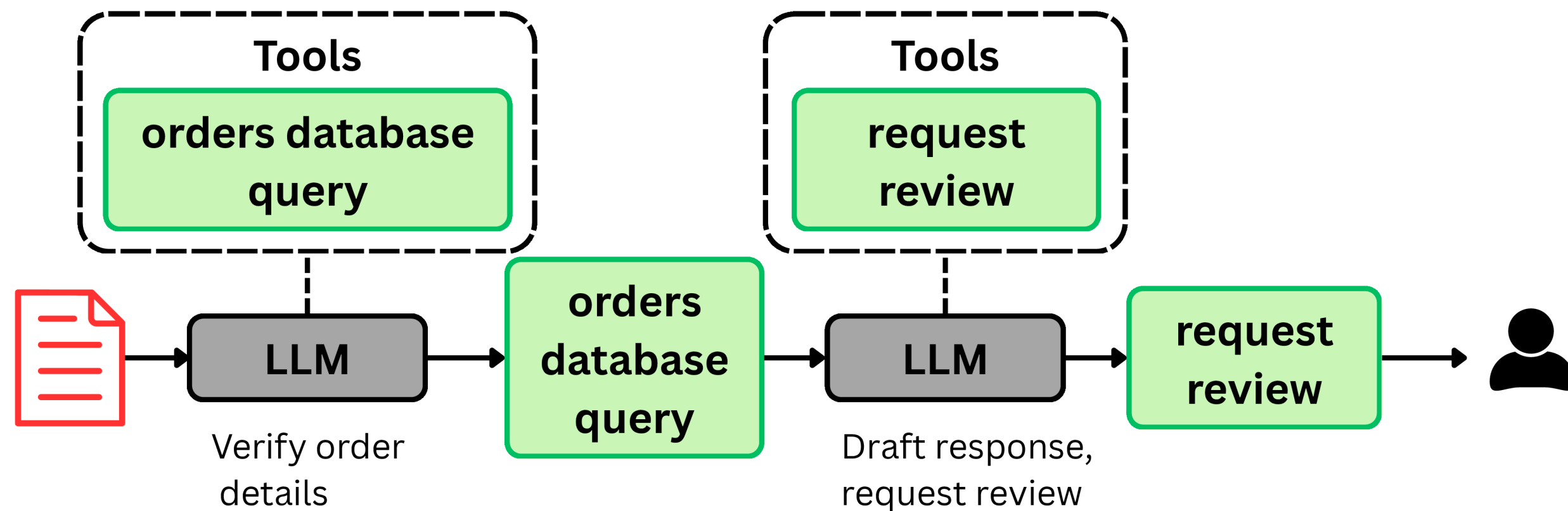
I ordered a blue KitchenPro blender (Order #8847) but received a red toaster instead.

I need the blender for my daughter's birthday party this weekend. Can you help?

Susan

Steps:

1. Extract key information
2. Find relevant customer records
3. Draft response for human review



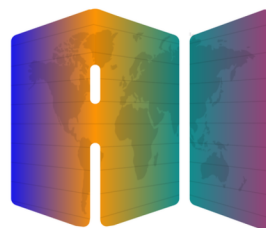
Counting up the errors

Input	LLM-drafted query	Orders database query	LLM-drafted email
Email 1	Wrong table		
Email 2		Error in database entry	Didn't address details of order
Email 3	Incorrect math		
...	...		...
Email 50			Defensive tone

75%

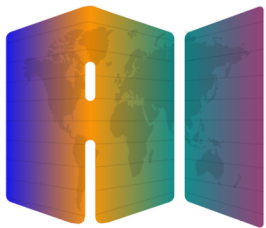
4%

30%

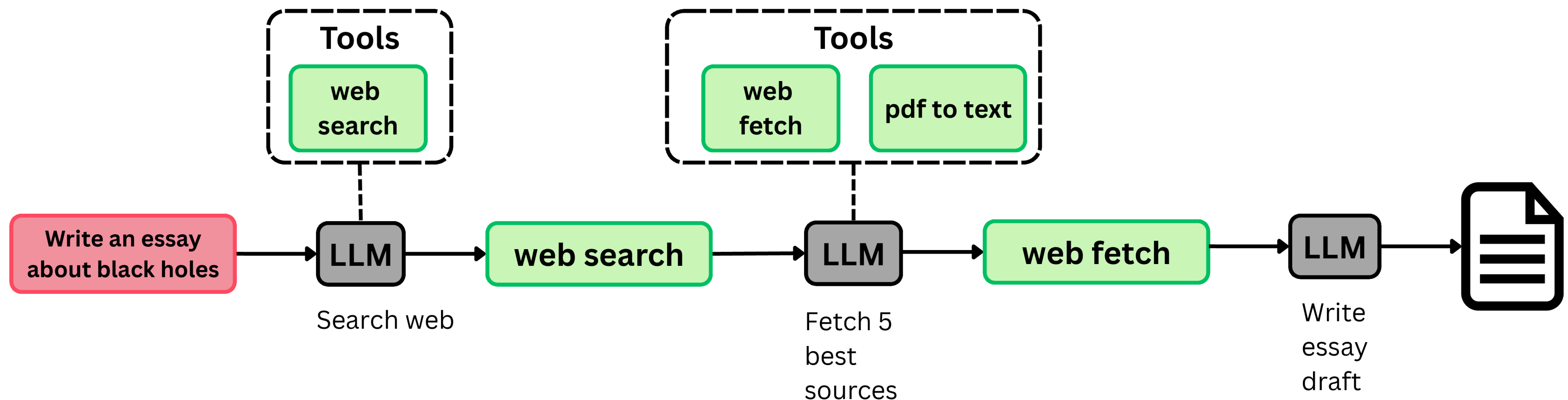


Practical Tips for Building AI Agents

Component-level evaluations

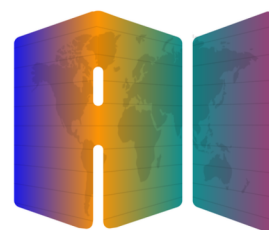


Example: research agent

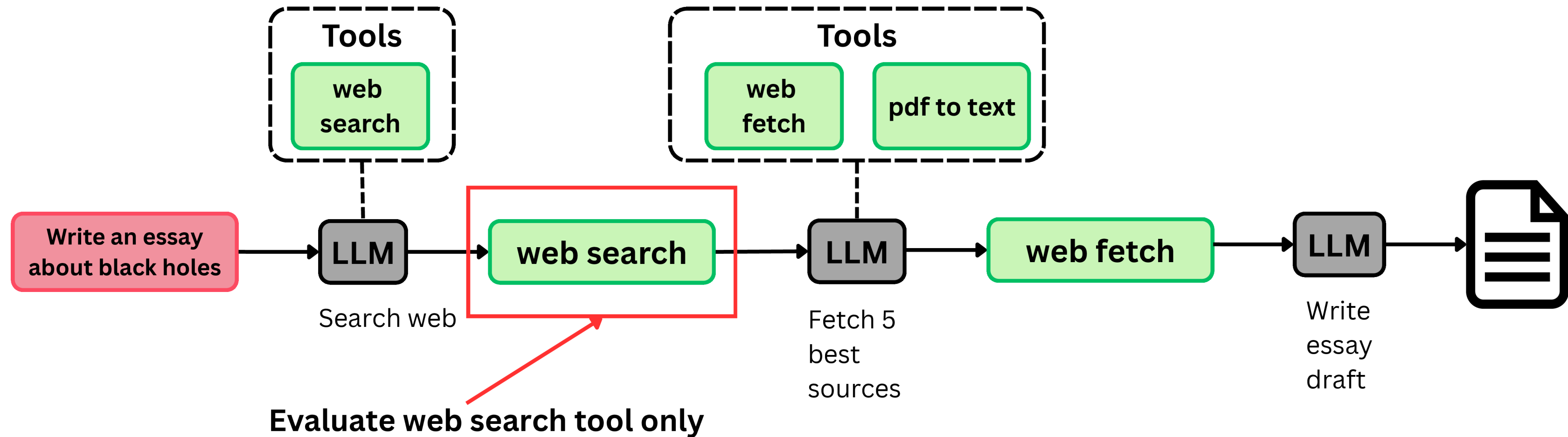


Propmt	Issues
Recent black hole science	Missed high-profile result that had lots of news coverage
Renting vs buying a home in Seattle?	Seems to do a good job
Robotics for harvesting fruit	Didn't mention leading equipment company

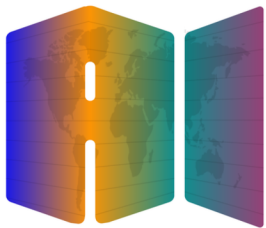
End-to-end eval is expensive



Example: research agent

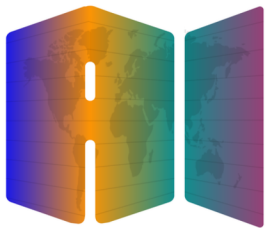


- Create a list of gold standard web resources
- Write code that calculate how many results correspond to gold standard websites e.g. F1-score
- Track as you vary hyperparameters: e.g., search engine, number of results, dates

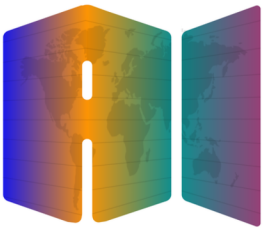


Benefits of component-level evaluations

- Can provide clearer signal for specific errors
 - Avoid the noise in end-to-end system
- More efficient for focused team to optimize
 - Work on a smaller, more targeted problems faster

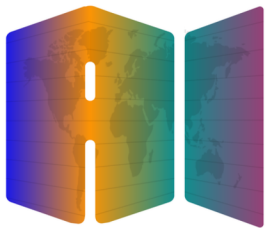


tryout-1.ipynb



Practical Tips for Building AI Agents

**How to address problems
you identify**



Improving non-LLM component performance

e.g. web search, text retrieval for RAG, code execution, trained ML model (for speech recognition, people detection, etc.)

- **Tune hyperparameters of component**

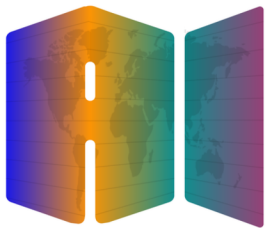
Web search: Number of results, date range

RAG: Change similarity threshold, chunk size

ML models: Detection threshold

- **Replace the component**

Try a different web search engine, RAG provider, etc



Improving LLM component performance

- **Improve your prompts**

Add more explicit instructions.

Add one or more concrete example to the prompt (few-shot prompting)

- **Try a new model**

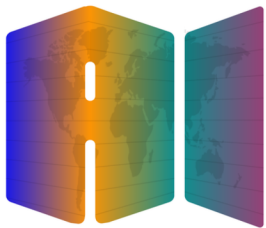
Try multiple LLMs and use evals to pick the best

- **Split up the step**

Decompose the task into smaller steps

- **Fine-tune a model**

Fine tune on your internal data to improve performance



Instruction following

Summary of customers call:

On July 14, 2023, Jessica Alvarez (SSN: 555-44-3333) of 1024 Maple Ridge Lane, Boulder, CO 80301, submitted a support ticket...

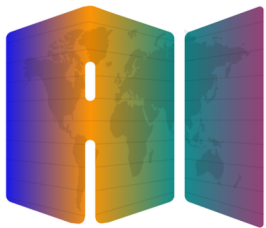
Prompt

Identify all cases of personally identifiable information (PII) in the text below.

Then return a list of the identified PII classified by type, and then redact all the identified PII with “*****”.

Separate the list and the redacted text with “REDACTED:”.

{text}



Instruction following results (Llama 3.1 8B)

The identified personally identifiable information (PII) classified by type is:

- Social Security Number (SSN): 524-18-7629
- Address: 1024 Maple Ridge Lane, Boulder, CO 80301

Redacted text with “*****” replacing the identified PII:

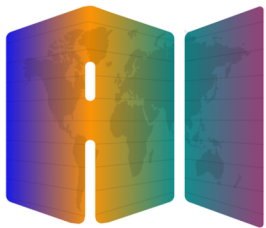
On July 14, 2023, ***** (SSN: *****) residing at *****, Boulder, CO 80301, submitted a support ticket...

List of identified PII classified by type

- SSN: 524-18-7629
- Address: 1024 Maple Ridge Lane, Boulder, CO 80301

Didn't follow the instructions properly

Missed some PII



Instruction following results (GPT-5)

Identified PII (type → value):

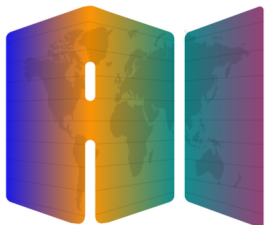
1. Full Name → Jessica T. Alvarez
2. Social Security Number → 524-18-7629
3. Physical Address → 1024 Maple Ridge Lane, Boulder, CO 80301

Followed formatting instructions

Identified all PII

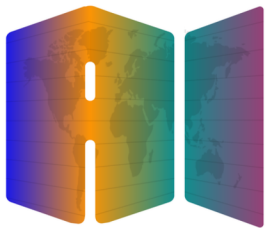
REDACTED:

On July 14, 2023, ***** (SSN: *****) residing at *****, submitted a support ticket...



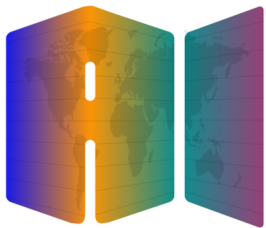
Developing intuition for model intelligence

- Play with models often
 - Having a personal set of evals might be helpful
 - Read other people's prompts for ideas of how to best use models

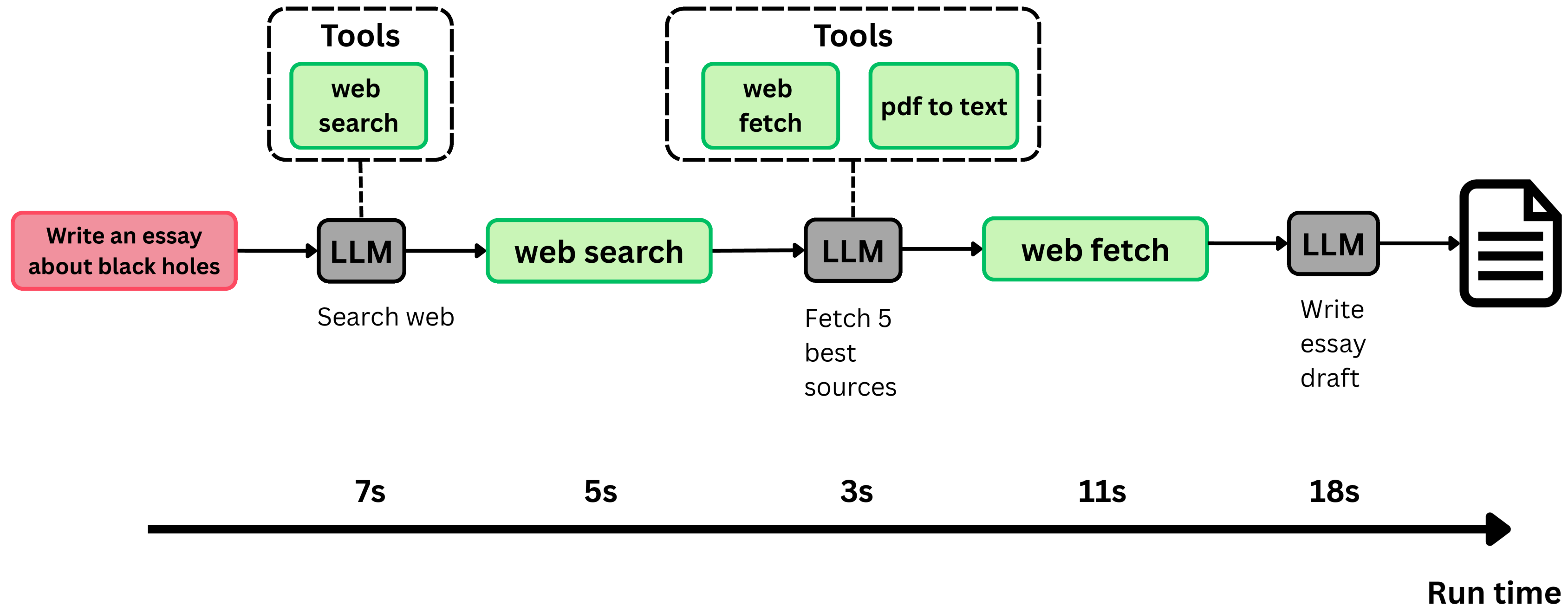


Practical Tips for Building AI Agents

Latency, cost optimization



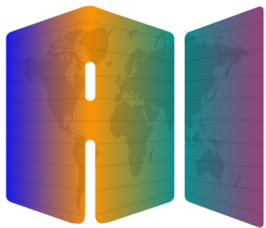
Example: research agent



Consider parallelism?

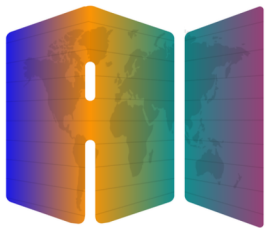
LLM steps too long?

- Try smaller/less intelligent model, or faster LLM provider

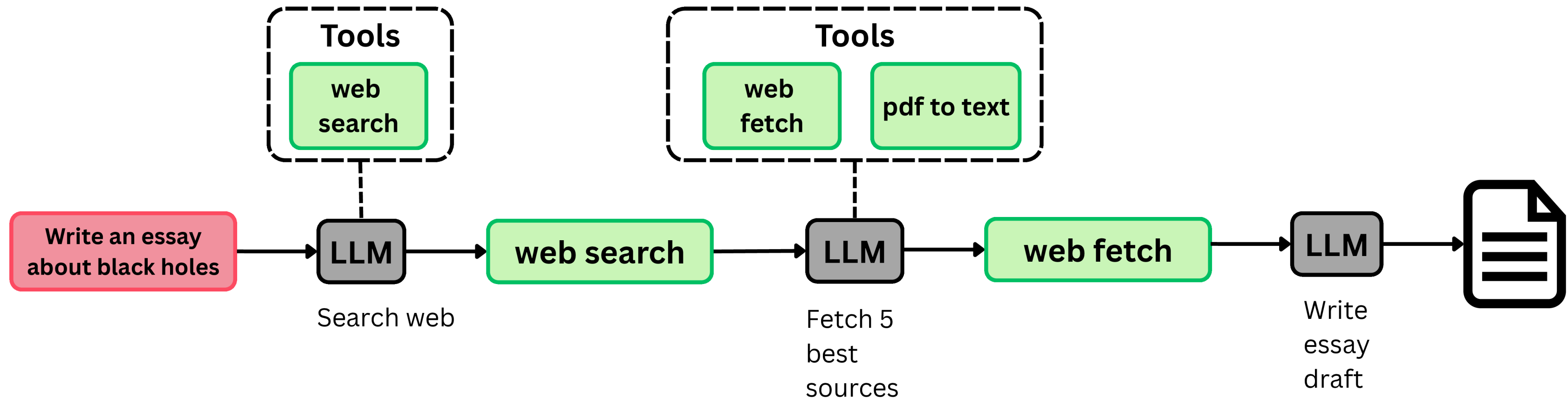


Costing your workflow

- LLM steps (pay per token)
- Any API-calling tools (pay per API call)
- Compute steps (based on server capacity/cost)



Costing your workflow



Tokens:
Php0.025

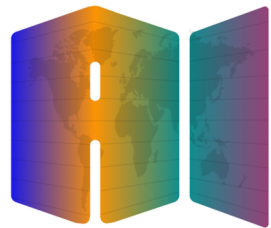
API call:
Php0.98

Tokens:
Php0.025

API call:
Php0.12

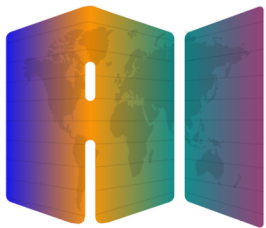
Tokens:
Php0.055

PDF-to-text:
Php1.85



Practical Tips for Building AI Agents

**Development process
summary**



Development process summary

